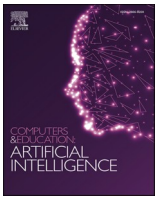
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Automatic item generation in various STEM subjects using large language model prompting

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ABSTRACT

Large language models (LLMs) that power chatbots such as ChatGPT have capabilities across numerous domains. Teachers and students have been increasingly using chatbots in science, technology, engineering, and mathematics (STEM) subjects in various ways, including for assessment purposes. However, there has been a lack of systematic investigation into LLMs' capabilities and limitations in automatically generating items for STEM subject assessments, especially given that LLMs can hallucinate and may risk promoting misconceptions and hindering conceptual understanding. To address this, we systematically investigated LLMs' conceptual understanding and quality of working in generating question and answer pairs across various STEM subjects. We used prompt engineering on GPT-3.5 and GPT-4 with three different approaches: standard prompting, standard prompting with added chain-of-thought prompting using worked examples with steps, and the chain-of-thought prompting with coding language. The questions and answer pairs were generated at the pre-university level in the three STEM subjects of chemistry, physics, and mathematics and evaluated by subject-matter experts. We found that LLMs generated quality questions when using the chain-of-thought prompting for both GPT-3.5 and GPT-4 and when using the chain-of-thought prompting with coding language for GPT-4 overall. However, there were varying patterns in generating multistep answers, with differences in final answer and intermediate step accuracy. An interesting finding was that the chain-of-thought prompting with coding language for GPT-4 significantly outperformed the other approaches in generating correct final answers while demonstrating moderate accuracy in generating multistep answers correctly. In addition, through qualitative analysis, we identified domain-specific prompting patterns across the three STEM subjects. We then discussed how our findings aligned with, contradicted, and contributed to the current body of knowledge on automatic item generation research using LLMs, and the implications for teachers using LLMs to generate STEM assessment items.

1. Introduction

Generative artificial intelligence (AI) is a relatively new form of content creation AI that has shown a lot of promise in many fields. Unlike previous forms of AI, generative AI is highly steerable by human prompts and very versatile and knowledgeable in many different domains, including the field of education (Mishra et al., 2023). An emerging number of studies have been examining the use of generative AI in teaching and learning (Ali et al., 2023; Lo, 2023). One of the possible applications of generative AI is to automatically create academic content, particularly for assessment purposes. Such an application would, among many benefits, reduce the cost and effort to implement assessment, improve scalability to benefit more students, and increase customization by personalizing assessment to individual students (Circi et al., 2023).

However, there is a need to systematically investigate the capabilities of generative AI, specifically large language models (LLMs), in automatic assessment item generation. This research need is especially important for science, technology, engineering, and mathematics (STEM) subjects for a number of reasons. First, assessments can be high stakes, necessitating a thorough investigation of the capabilities and limits of LLMs. This issue is crucially important given LLMs' tendency to hallucinate (confabulate), since by design they are 'forced' to autocomplete text (Xu, Jain, & Kankanhalli, 2024). Second, STEM subjects,

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especially at advanced levels, rely on the precision of scientific and mathematical computations that LLMs may not be optimized or fine-tuned for, an issue that has been rarely systematically investigated. Finally, one of the major capabilities of LLMs has been its ability to learn in context via human prompts without needing additional neural network training (Brown et al., 2020). However, there have been many different prompting strategies, and determining which ones are optimal for STEM automatic assessment item generation has been unclear. To date, there has been a lack of systematic research on whether and how LLMs can be used for automatic item generation across STEM subjects. For these reasons, our study is guided by the following research question: *How well do LLMs generate questions and answers in various STEM subjects in terms of conceptual accuracy and quality of working?*

2. Literature review

2.1. Automatic item generation in education

Automatic item generation (AIG; also termed 'automated item generation' or 'automatic question generation') has been increasingly gaining attention in educational assessment research with the development of machine learning, natural language processing, and large language model (LLM) technologies (Gierl et al., 2022; Kurdi et al., 2020). This interest has arisen from the need for effective ways of generating items including questions and answers at lower costs and in less time and the need for supporting students' learning through customized and tailored measurement items (Circi et al., 2023). Generating items or questions tends to require significant effort from teachers, as formative and summative assessments are generally required in various educational contexts. To serve this demand and allow teachers to focus on more essential areas, considerable effort has been made to establish systems with better capabilities for automatic item and question generation. In addition, teachers being able to use readily available, high-quality items to assess their students tends to increase the opportunity to provide more appropriate instruction based on quality evaluation. With these reasons in mind, there have been some studies on AIG in education, including in STEM subjects. STEM disciplines such as chemistry, physics, and mathematics have traditionally incorporated problem-solving activities into instructional approaches (Gabel et al., 1984; Larkin & Reif, 1979; Lesh, 1981), as solving quantitative problems may enhance students' coherent conceptual understanding by involving significant practices in the use of equations and inferences (e.g., McDermott, 2001). For example, efforts have been made to establish a larger bank of physics test items using AIG (Mindyarto et al., 2018).

One of the key issues in AIG studies has been securing higher quality items while ensuring their scalability (Gierl & Haladyna, 2012). Significant efforts have been made to ensure consistency in the high quality of automatically generated items, resulting in item quality that is comparable between AIG and manual item generation (Falcão et al., 2023). Various efforts have also focused on ensuring robust methodologies in terms of validity, reliability, and item discrimination and difficulty in AIG (Tan et al., 2024). For instance, ensuring construct validity involves developing algorithms that generate items aligning with theoretical constructs and validating them empirically through statistical methods such as factor analysis and correlations with other measures (Gierl et al., 2022). Furthermore, continuous development in AIG studies has been accelerated by advancements in AI technologies, particularly generative AI, which have broadened the potential for innovation in this area.

On the other hand, AIG can generally be categorized into two input types: (1) structured input and (2) unstructured input (Circi et al., 2023). *Structured input* refers to using item models, which are predefined templates to guide item generation. These templates provide a clear framework that outlines the specific components or logical steps of questions, ensuring consistency and alignment with standards. This approach is particularly useful for standardized testing, where uniformity and reliability are crucial (Gierl & Lai, 2013). For example, an item model template for a physics question might specify the type of problem, the variables involved, and the required operations, enabling the AIG system to generate a variety of questions that fit the predefined criteria. Structured input not only enhances the efficiency of item generation but also maintains the quality and validity of the items produced. *Unstructured input* refers to generating assessment items by processing natural language to understand the data given without any predetermined templates. This type is particularly useful in generating items requiring higher-level skills. For example, to generate reading comprehension questions based on a passage from a novel or an article, the AIG system may craft questions probing the reader's understanding of themes, characters, and events (Attali et al., 2022). However, challenges exist in ensuring the consistency and quality of the generated items, necessitating robust validation and review processes to maintain assessment standards (Liu et al., 2021). There can also be cases of combining both structured and unstructured inputs, which depends on the purpose of item generation, the practical context of AIG, and their respective pros and cons.

2.2. Large language models in education and for AIG

Along with the advancement of machine learning and natural language processing technologies and the use of large amounts of public digital data for training, LLMs, as one of the emerging generative AI technologies, have recently been regarded as very significant and indispensable tools in shaping the future of education, based on the shared public recognition of their considerable potential benefits (Grassini, 2023). Due to LLMs' ability to interact using human-like language (Ahmad et al., 2023), there have been various revolutions in educational contexts, such as developing teaching materials (Küchemann et al., 2023), affording personalized learning experiences (Halaweh, 2023), providing immediate feedback (Meyer et al., 2024), reducing teachers' workload (Limna et al., 2023), supporting idea generation (Thanasi-Boçe & Hoxha, 2024), assisting with writing tasks (Stahl et al., 2024), and evaluating ideas (Xu et al., 2024). However, there have also been challenges in using LLMs, including responding to biased perspectives (Zhang & Tur, 2024), dealing with hallucinations (Mishra et al., 2023), and addressing integrity issues (Cotton et al., 2024). As this area of generative AI in education is emerging, a variety of opportunities and challenges in theoretical, practical, and technological aspects have been newly suggested and discussed (Chiu et al., 2023).

One of the exciting areas in educational research involving LLMs is AIG. We illustrate the benefits of the use of LLMs for AIG to create quality assessment items and discuss the use of prompt engineering techniques to enhance coherence and address their challenges. In addition, we introduce a coding language approach that can be integrated into prompt engineering embedded in LLMs.

2.2.1. Benefits of LLMs for AIG

LLMs have the potential to play a transformative role in AIG research, resulting in significant developments in the quality of generated items and questions, the efficiency of item generation, and the accessibility of LLMs. First, there have been significant improvements in the quality of questions generated by LLMs, which have become comparable to manually generated items (Kıyak & Emekli, 2024). LLMs leverage vast amounts of pre-training data, enabling them to generate high-quality items based on contextual understanding. With the development of more advanced LLMs, such as Generative Pre-Trained Transformer 4 (GPT-4) and Llama 2, the capabilities have been reliably enhanced in terms of nuanced conversation. Specifically, GPT-4 has generated multiple-choice questions (MCQs) with qualities better than those produced by undergraduate and graduate students (Säuberli & Clematide, 2024).

Second, the efficiency of AIG has been considerably improved by LLMs. The most prominent feature of using LLMs for AIG has been the reduction in time and effort required for item generation, allowing

teachers to focus on more important tasks, such as interacting with their students. According to Cheung et al.'s (2024) study comparing ChatGPT with human professorial staff, the time taken to generate MCQs by humans was about ten times longer than by ChatGPT, while the quality of the questions was comparable. This critical feature also increases the scalability of AIG, equipping teachers with a vast pool of questions, which is one of AIG's major goals (Kurdi et al., 2020).

Third, many LLMs with natural language processing technologies allow human-like conversations more comprehensively, using their own language involving multimodal resources in AIG (Xie et al., 2024). This greatly enhances their accessibility and adaptability to non-computer language experts. Recently, there have been considerable studies involving LLMs by researchers and teachers for generating multimodal resources for teaching and learning without the use of programming languages such as Python and C++. Although there have been efforts to develop better prompting to yield outcomes from the LLMs, it has become much easier to automatically generate items/questions and their answers, along with improved generation quality and efficiency. Given that the utilization of LLMs in educational contexts is at a nascent stage, we may see further investigations, including AIG. With the enhanced usability of LLMs, more educators have easier access to them, which is one of the key motivations behind our study. In sum, these three primary benefits of using LLMs for AIG (quality, efficiency, and accessibility/adaptability), among various others, can be further strengthened given their great potential and increasing interest in them in educational fields.

2.2.2. Current issues challenged in AIG by LLMs

AIG research involving LLMs being in an early stage means that there have been various challenges, of which we will highlight two important aspects: ensuring the generation of high-quality questions and the adequacy of content knowledge in the generated items and answers. This section addresses these two challenging issues, which constitute the research gap addressed in this study.

2.2.2.1. Reliably ensuring high quality of items. Generating high-quality items reliably has been a primary goal of AIG. The current research trend has focused more on generating MCQs (e.g., Kiyak et al., 2024), which employ structured input through item models or item templates, than on generating questions with unstructured input, which require more contextual responses from LLMs. Although many studies have reported that automatically generated items have been comparable in quality to those created by humans (Cheung et al., 2023), this quality challenge remains when we need more rigorous assessments. For example, Artsi et al. (2024) reviewed eight empirical studies of medical education examining the quality of MCQs generated by either GPT-3.5 or GPT-4 and in all studies found faulty questions that were inappropriate for exams. Since the structured input approach for MCQ generation that was applied in the studies can be more reliable compared to unstructured input, there is a further need for the development of prompt engineering to obtain more reliable and accurate AIG, despite significant improvements in the field.

Few-shot and chain-of-thought approaches for AIG. One major finding from previous studies investigating prompt engineering for the use of LLMs has been the importance of developing quality prompting (Kıyak, 2024). The quality of items is largely and sensitively affected by the way prompting is done. Considering this, there have been examinations to find optimal promptings using the few-shot approach (e.g., Lee et al., 2023; Lee et al., 2024) and the chain-of-thought (CoT) prompting (Wei et al., 2022), which have outperformed zero-shot and non-CoT approaches, respectively. The *few-shot* approach involves providing examples for generation, whereas zero-shot prompting only states the requirement to be generated. For example, providing a sample question and answer may guide the logic of the questions and answers generated by LLMs. The *CoT* approach guides response generation through multiple steps to elicit reasoning steps, significantly reducing errors and hallucinations (Wei et al., 2022). For example, informing LLMs with multiple steps in prompting and instructing them to follow these steps is an example of CoT. The combination of standard prompting (SP) and the CoT approach has become popular in recent studies (e.g., Cohn et al., 2024; Lee et al., 2024). Although there has been an increase in studies in educational assessment, there has been a lack of empirical research investigating LLMs for AIG, particularly in STEM disciplines employing these approaches. This study addresses one of these research gaps.

Coding language embedded in LLMs for AIG. One novel approach in prompt engineering to ensure and improve the quality of response generation is integrating a coding language into prompting of LLMs. This approach involves emulating the multiple steps provided by a CoT approach within a separate window, allowing for the verification of steps apart from the internal generation process in LLMs. In particular, this function enables translating natural language containing logical structures to coding language (e.g., Python) immediately by an iterative process, including building the algorithm and testing and modifying it. The coding language converted from natural language reduces ambiguity, as general natural language can be understood in multiple ways that may result in miscommunication. One of the challenges of using coding language is the demand for technical knowledge, making its accessibility low (Connolly, 2001). However, with advancements in LLMs such as GPT-4, this accessibility challenge has been considerably reduced (Shipps, 2024), allowing us to straightforwardly generate coding language from simple natural language for AIG. As this is a new approach, there have been very few studies involving the integration of a coding language approach into prompting in LLMs for AIG. We believe that this novel attempt may contribute to enhancing item quality, as STEM subject questions often involve highly logical and multistep processes to be solved in a closed-ended manner, something that coding achieves well.

In sum, given the current challenging issues and the approaches discussed, we investigated how SP, CoT, and coding approaches generate quality questions and their answers. Additionally, to understand how advancements in LLMs have an impact on quality, we compared a previous model, GPT-3.5, with a recent model, GPT-4.

2.2.2.2. Domain-specific knowledge generation capabilities. One important question when prompting item generation using LLMs is how well their answers align with current domain-specific knowledge. According to OpenAI's (2023) technical report, GPT-4 demonstrates human-level performance on various professional and academic examinations. However, there has been a persistent question regarding the capabilities of LLMs in terms of conceptual understanding in different disciplines. Recently, Balhorn et al. (2024) conducted an intriguing investigation into how well LLMs answer questions in the natural sciences and engineering. The results revealed that the average quality of answers generated by LLMs was mostly correct at the undergraduate level, as judged by experts in each discipline, but only partly correct at the master's and PhD levels. Although the investigation was carried out using a zero-shot approach, the results might be promising if we ask LLMs at the undergraduate level. ChatGPT is a pre-trained model using digital data with a cut-off in September 2021, and a more recent model was trained on more recent public data, which may cover content in K-12 STEM subjects in various national curricula, so it is likely there would be fewer problems at the secondary school level.

On the other hand, one of the findings in the area of LLM research is that LLMs have performed differently depending on domain-specific knowledge (OpenAI, 2023). Although GPT-4 outperformed previous models such as GPT-3.5, the results revealed that the accuracy of factual knowledge evaluation varied across different domains, such as mathematics and science. Although there have been studies investigating LLMs' capabilities within particular STEM subjects such as physics (Wancham et al., 2023), chemistry (Moore et al., 2024), and

mathematics (Feng et al., 2024), it has been rare to systematically investigate the automatically generated items across different subjects, which is another research gap. In addition, given that MCQs have been shown to be limited in fully evaluating students' conceptual understanding (Brassil & Couch, 2019), it is necessary to explore other types of questions, such as problems requiring multistep answers. It is critically important to incorporate problem-solving activities in science, for example, in chemistry and physics and mathematics lessons (Taconis et al., 2001), to give students opportunities to improve their conceptual understanding by engaging in problem-solving tasks that involve both conceptual reasoning and procedural knowledge (Hiebert, 1986; McDermott, 2001). Considering this gap, we aimed to investigate how LLMs generate quality questions and answers across different STEM subjects, in particular, chemistry, physics and mathematics, at the pre-university level in terms of conceptual accuracy and quality of working.

3. Research questions

Based on the literature review, we formulated the following specific research questions to investigate how well LLMs generate questions and answers in the STEM subjects of chemistry, physics, and mathematics at pre-university levels.

- RQ1: How well do GPT-3.5 and GPT-4 perform in generating questions and answers in terms of conceptual accuracy and quality of working?
- RQ2: How well do SP, CoT prompting, and CoT with coding language prompting approaches perform in generating questions and answers in terms of conceptual accuracy and quality of working?
- RQ3: How do LLMs differ in generating questions and answers across different STEM subjects in terms of conceptual accuracy and quality of working?
- RQ4: What are the common features of errors found in questions and answers generated by LLMs?

4. Methods

This section outlines methods for generating items, including questions and answers, using LLMs and evaluating their accuracy and quality across different LLMs, prompting approaches, and three different STEM subjects at the pre-university level. This process had four phases: data collection and data analysis. The initial phase involved designing the experiments for the study and developing sample questions and answers for specific topics in three subjects: chemistry, physics, and mathematics. These sample questions and answers served as references for automatically generating items. The second phase involved developing three prompting approaches: (1) SP, (2) CoT, and (3) CoT with coding language (CoT-CL). How these prompting approaches were developed is illustrated with examples in Section 4.2 and explained in more detail in the Appendix. The third phase involved evaluating the questions and answers in terms of a rubric developed. For this, we developed a rubric consisting of four aspects: (a) clarity and (b) context accuracy of questions, (c) final answer accuracy, and (d) quality of working for the answers. The last phase is data analysis, which involved descriptive statistics, analysis of variance (ANOVA), and case specific analysis for each topic.

4.1. Experimental design and sample item development

As discussed in the literature review section, this study compared how well GPT-3.5 and GPT-4 generated questions and answers across three different STEM subjects in terms of accuracy and quality. In addition, this study also compared three prompting approaches: SP, CoT, and CoT-CL. For this, we generated 600 questions and answer pairs using GPT-3.5 and GPT-4 across three subjects, as shown in Table 1. The detailed design is illustrated in the following section.

The sample question and answer pairs were developed for three subjects at the Higher 2 level of the Cambridge International General Certificate of Education: Advanced Level syllabus: chemistry, physics, and mathematics. Within each subject, two different topics were selected to ensure a wider range of scopes, and one question for each topic was developed involving quantitative calculation (see Table 2). The selected topics and questions, which require multistep answers including calculations, are typical of high-stakes examinations and are frequently used in pre-university classes for the three subjects. Six co-authors who were subject-matter experts—two from each subject—and who teach at the pre-university level developed the sample questions and answers for reference in generating similar items using LLMs. The developed questions and answers were generally equivalent to the learning outcomes of the respective Higher 2 Advanced Level syllabuses. The developed sample items were used only for the CoT and CoT-CL approaches, while the SP approach involved question and answer generation using GPT-3.5 and GPT-4 without any sample items.

4.2. Development of prompting approaches

Following the selection of the six topics, three prompting approaches, as shown in Table 3, were developed and employed: (1) The

Table 2
Topics selected as references for automatic item generation.

Subject	Topic	Description
Chemistry	Titration	Back-titration of insoluble carbonates
	Electrolysis	Mass of deposition of metal through electrolysis
Physics	Projectile motion	Kinematics: coordination of the motions (e.g., having a depression, moving in horizontal uniform circular motion)
	Circular motion	Conical pendulum moving in horizontal uniform circular motion
Mathematics	System of linear equations	System of linear equations with three variables in real-world context
	Permutation and combination	Number of ways of choosing r distinct objects out of n distinct objects, when order matters and when order does not matter

Table 1
Dataset for experiment.

	Prompting approach	Chemistry		Physics		Mathematics		Total
		Topic 1	Topic 2	Topic 1	Topic 2	Topic 1	Topic 2	
GPT-3.5	SP	20	20	20	20	20	20	120
	CoT	20	20	20	20	20	20	120
	Subtotal	40	40	40	40	40	40	240
GPT-4	SP	20	20	20	20	20	20	120
	CoT	20	20	20	20	20	20	120
	CoT-CL	20	20	20	20	20	20	120
	Subtotal	60	60	60	60	60	60	360
Total		100	100	100	100	100	100	600

Table 3
Descriptions of three prompting approaches for this study.

Prompting approach	Description	Prompt example [*note*]
SP	Prompt to generate questions and answers with contextual information, such as subject, topic, level of teaching, and context of the question	You are a physics [*subject*] teacher designing practice questions for Year 12 students [*level of teaching*]. You will be designing a calculation question for practice [*context of the question*]. Please create a kinematics question for a ball that is kicked upwards at an angle and lands into a depression [*topic*]. Please provide the question and answer scheme.
CoT	Build on the SP, prompt to create a sample question and answer, and step-by-step reasoning	Here is a sample question and answer, which you will follow closely to create new questions and answers [*step-by-step reasoning*]. – Sample Question [*example*]: A ball was kicked with a velocity of 25 m s^-1 at an angle of 50° above the horizontal. It landed into a depression 2.0 m deep. Calculate the velocity of the ball just before it hit the depression. – Sample Answer [*example*]: vy^2 = uy^2 + 2aysy 0 = (usin(50°)) + 2*(-9.81)*(8.0) u = 16.35 m/s vy^2 = (uy^2) + 2aysy vy^2 = ((25sin(50°))^2) + 2*(-9.81)*(-2.0) vy = 20.150 m/s vx = 25*cos(50°) = 16.070 m/s v^2 = vx^2 + vy^2 v = sqrt((16.070)^2 + (20.150)^2) = 25.8 m/s tan(theta) = vy/vx = 20.150/16.070 theta = 51.4° below horizontal
CoT-CL	Build on the CoT, prompt to write code to emulate the work, including an additional verification step	Please write Python code to emulate the mark scheme and run the code to ensure that you get the same answer [*coding language*]. Otherwise, rewrite and rerun the code [*verification*].

SP approach provided contextual information for each topic, such as the subject, level of teaching, and the context of the question, to enhance the quality of generation (Liu et al., 2023). (2) The CoT approach in this study integrated a few-shot technique (Lee et al., 2023) with step-by-step reasoning (Wei et al., 2022) by providing a prompt to follow the steps with a sample question and answer pair building on the SP approach. (3) The CoT-CL approach further added an instruction to run code in the prompt, expanding on the CoT approach. Using these three different approaches, 20 question-and-answer pairs were repeatedly generated for the three subjects using GPT-3.5 and GPT-4. However, due to the absence of a code interpreter function in GPT-3.5, the CoT-CL approach was employed only with GPT-4 (see Table 1). Detailed promptings for each approach are illustrated in the following subsections.

To develop the three prompting approaches, we conducted an iterative process of drafting prompts, reviewing the generated results, revising, and validating the prompts, guided by Lee et al. (2024). During this process, we identified subject- or topic-specific rules for prompting and included them differently across various topics. For example, prompting the use of the same chemicals in a chemistry question made the outputs more accurate. To establish the validity of the prompts and a common understanding, five question and answer pairs for each prompting approach across three subjects were drafted, independently reviewed, revised, and validated by the authors. This process involved subject-matter experts from the respective subjects and experts in AI in education for further data collection. After reaching agreement, 20 question and answer pairs per prompting approach for each topic were automatically generated by GPT-3.5 and GPT-4.

4.2.1. SP

The SP technique involves the human prompter stating the subject, level of teaching, the context of the question, and the requirement of creating a question along with the answer scheme. Once LLMs provided the question and answer scheme by prompting 'Please provide the question and answer scheme', they were used as the sample items to generate another 19 questions and answers. This additional line 'Please generate simple text that I can copy easily' was also practically useful to easier transfer of results into other sheets.

> You are a <subject> teacher designing practice questions for year 12 students.
> **You will be designing a calculation question for practice**.
> Please create a question on < insert context>.
> Please provide the question and answer scheme. Please generate simple text that I can copy easily.

4.2.2. CoT

The CoT prompting technique builds on the SP approach, which provides contextual information and additionally provides the sample question and answer pair (few-shot technique) for the LLMs to refer to with a step-by-step reasoning prompt. For CoT, GPT-4 tends to run code even without any relevant prompt, so we added 'Do not generate or run code' to ensure the CoT approach is enforced. In addition, delimiters were found to be effective in separating the various sections for better clarity for the large language models (Bsharat et al., 2023). Hence, delimiters were used to separate the instruction, sample question, sample answer and output expected.

> You are a <subject> teacher designing practice questions for Year 12 students.
> You will be designing a calculation question for practice.
> **Here is a sample question and answer, which you will follow closely to create new questions and answers.**
> –
> Sample Question:
> <insert sample question here>
> –
> Sample Answer:
> <insert sample answer here>
> –
> Do not generate or run code.
> Please generate simple text that I can copy easily.

4.2.3. CoT-CL

Building on CoT, CoT-CL added a prompt to write code to emulate the work. The generated item was then validated using the Code Interpreter function in GPT-4 to verify if the code was correct. If the code was

accurate, further questions and answers could be written.

You are a <subject> teacher designing practice questions for Year 12 students.
You will be designing a calculation question for practice.
Here is a sample question and answer, which you will follow closely.
–
Sample Question:
<insert sample question here>
–
Sample Answer:
<insert sample answer here>
–
Please write Python code to emulate the mark scheme and run the code to ensure that you get the same answer. Otherwise, rewrite and rerun the code
Please generate simple text that I can copy easily.

4.3. Evaluation of generated question and answer pairs

To evaluate the generated questions and answers, we developed a rubric that included two aspects for the quality of the questions and another two aspects for the quality of the answers: (1) clarity and (2) context accuracy for questions and (3) final answer accuracy and (4) quality of working for answers (see Table 4). These components were extracted based on the literature on test development (Lane et al., 2016) and AIG research (Kurdi et al., 2020). *Clarity* of the question refers to whether the generated question is clear in terms of readability, language clarity, and logical order of information. *Context accuracy* of the question refers to whether the context of the questions is realistically and factually appropriate. *Final answer accuracy* pertains to the correctness of the final answer generated and does not account for the intermediate steps. *Quality of working* in answers refers to whether the multistep answers are correct and the necessary steps are provided and not redundant. Scoring was either 0, 1, or 2 depending on the meeting the rubrics (see Table 5). However, the difficulty (Gierl et al., 2022) aspect was not included since this study generated similar questions and answers from the initial or sample question and answer, resulting in very similar difficulty levels.

There were a total of three pairs of inter-rater reliability, one for each of chemistry, physics, and mathematics. The overall value of Cohen's kappa coefficient was 94%. Cohen's kappa coefficient of each subject was 93%, 95%, and 95% for chemistry, physics, and mathematics, respectively, indicating high inter-rater reliability (Wongpakaran et al., 2013). The following steps were performed to ensure inter-rater reliability. First, for each prompting approach, two subject-matter experts of the same subject co-evaluated the five preliminarily generated question-and-answer pairs to establish a common ground for scoring. Second, each rater independently evaluated the first 10 questions and answers in the dataset and then discussed their evaluations to ensure alignment of ideas. Once aligned, each rater evaluated half of the remaining 10 questions independently.

4.4. Data analysis

The dataset was analyzed at three different levels to address RQ1, RQ2, and RQ3: (1) two different LLMs (GPT-3.5 and GPT-4), (2) three different prompting approaches (SP, CoT, and CoT-CL), and (3) three different STEM subjects (chemistry, physics, and mathematics), focusing on clarity, context accuracy, final answer accuracy, and quality of working. The comparison between GPT-3.5 and GPT-4 was conducted using descriptive statistics, incorporating the three different prompting approaches across all subjects. To compare the three prompting approaches and the three STEM subjects, we used a one-way analysis of variance (ANOVA). To address RQ4, subject-matter experts evaluated all generated questions and answers in terms of clarity, context accuracy, final answer accuracy, and quality of working. This qualitative evaluation included comments on whether the questions and answers were clear and what errors were found. Based on the qualitative analysis, we identified disconnections between the intermediate steps and the final answer, as well as domain-specific errors across the three subjects. The qualitative analysis was conducted by the first three authors, and the process was repeated until agreement was reached. The purpose of this qualitative analysis was not to find universal patterns with probability but to illustrate specific, interesting patterns found in this study.

Table 4
Rubric for evaluation.

	Aspect	Question
Question	Clarity	● Is the question understandable? ● Is the language clear? ● Is the ordering of information logical?
	Context accuracy	● Is the context factually appropriate?
Answer	Final answer accuracy	● Is the final answer correct?
	Quality of working	● Are the steps correct? ● Are there redundant steps?

Table 5
Scoring questions and answers.

Score	Description
0	Poor, does not meet rubric requirement at all
1	Acceptable but can be improved/minor issues
2	Excellent, meets/exceeds rubric requirement

5. Findings

5.1. Quantitative analysis

5.1.1. Comparison between GPT-3.5 and GPT-4

GPT-4 outperformed GPT-3.5 in generating answers in terms of final answer accuracy and quality of working. However, there were not many differences in quality when generating questions in terms of clarity and context accuracy, as shown in Fig. 1. The use of the CoT approach made the results comparable to the quality of AIG between GPT-3.5 and GPT-4 in generating questions. Despite using the CoT approach in GPT-3.5, final answer accuracy and quality of working for the answers were lower, scoring 0.60 and 1.19 (out of 2), respectively. In contrast, the CoT approach in GPT-4 performed better than in GPT-3.5. The following sections focus on the analysis results of GPT-4.

5.1.2. Comparison across three prompting approaches for GPT-4

We found that all three prompting approaches generally generated quality questions, while CoT-CL significantly outperformed the other two prompting approaches in generating answers (see Fig. 2). All approaches (SP, CoT, and CoT-CL) demonstrated good clarity of questions (Table 6). However, CoT and CoT-CL generated significantly higher scores for context accuracy, 1.8125 and 1.7250, respectively, compared to SP (1.1292) at $p < .001$ level. On the other hand, the final answer accuracy for both SP and CoT had a low score of 0.9958, while CoT-CL (1.8458) demonstrated significantly higher scores than the other two at $p < .001$ level. For the quality of working for the multistep answers, CoT-CL (1.4292) underperformed compared to CoT (1.6667) significantly at the 0.05 level but outperformed SP (1.2167) significantly at $p < .01$ level.

5.1.3. Comparison across three STEM subjects for GPT-4

We found that the quality of questions generated by GPT-4 was generally good, with clarity being rated particularly high across all three subjects (see Table 7 and Fig. 3). The mean scores for clarity were 1.9125, 1.9917, and 1.8417 out of 2 for chemistry, physics, and mathematics, respectively. The scores for context accuracy showed a moderate level of quality across the three subjects, with acceptable scores of 1.4667, 1.6083, and 1.5917 for chemistry, physics, and mathematics, respectively. However, the quality of answers revealed significant differences across the three subjects (see Table 7 and Fig. 3). For final

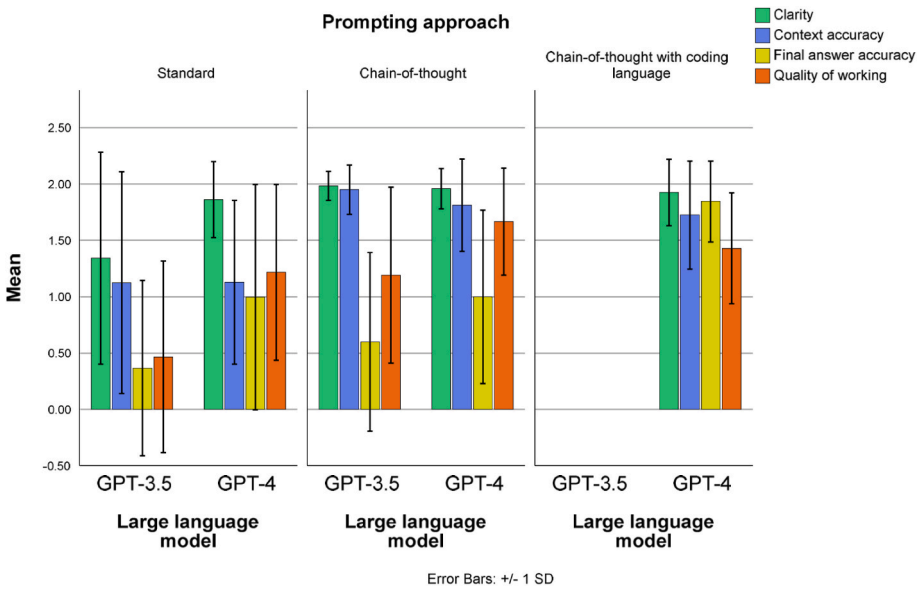


Fig. 1. Evaluation comparison between GPT-3.5 and GPT-4.

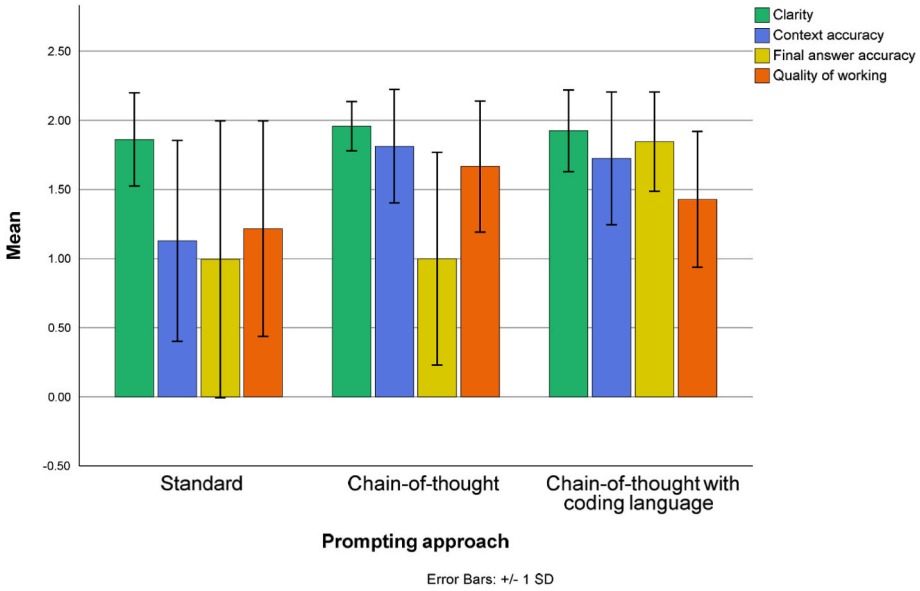


Fig. 2. Comparison across three prompting approaches for GPT-4.

answer accuracy, the chemistry score was significantly higher, followed by the mathematics and physics scores, with differences significant at the $p < .001$ level. Final answer accuracy for physics was the lowest among the three subjects at the $p < .001$ level, which might not be acceptable compared to chemistry and mathematics. For quality of working in multistep answers, the mathematics score (1.9792) was significantly higher than the other subjects, indicating very high quality. The scores for chemistry (1.1167) and physics (1.2167), however, were merely acceptable and might require improvement.

5.2. Qualitative analysis

We conducted qualitative analysis, focusing on the unsuccessful and noteworthy cases identified through the qualitative evaluation. Errors were classified into two broad categories: (1) errors found in the intermediate steps of multistep calculation answers or in the final answers and (2) domain-specific errors with common patterns. They are illustrated with several examples from three subjects in this section.

5.2.1. Unidentifiable connection between intermediate steps and the final answer

Overall, the quality of question generation was generally substantial in terms of clarity and contextual accuracy (see Table 6). However, multistep answer generation scored relatively lower and exhibited various error patterns. One pattern revealed by the qualitative analysis was that, in many cases where GPT-4 used the CoT approach, quality of working involving multistep equations to solve the questions was good, but the calculations in the intermediate and/or final answers were incorrect. Cases 1 and 2 (Table 8, Physics Topic 1 and Mathematics Topic 1) are the exemplary cases. These cases also help explain why the score for final answer accuracy was lower than that for quality of working (see Table 6). Another pattern we found was that final answer accuracy and quality of working seemed irrelevant to each other in some cases. For instance, incorrect intermediate steps might lead to either a

Table 6
One-way ANOVA of clarity, context accuracy, final answer accuracy and quality of working across type of prompting approach for GPT-4.

Variable	Prompt type (n = 120)	Mean	SD	F (p)	Scheffe
Clarity	SP[a]	1.8625	0.33658	3.674	a < b*
	CoT[b]	1.9583	0.17850	(0.026)	
	CoT-CL[c]	1.9250	0.29456		
Context accuracy	SP[a]	1.1292	0.72615	53.750	a < b***
	CoT[b]	1.8125	0.41028	(<0.001)	
	CoT-CL[c]	1.7250	0.48006		a < c***
Final answer accuracy	SP[a]	0.9958	1.00104	50.047	b < c***
	CoT[b]	1.0000	0.76970	(<0.001)	
	CoT-CL[c]	1.8458	0.35971		a < c***
Quality of working	SP[a]	1.2167	0.77982	17.003	c < b**
	CoT[b]	1.6667	0.47338	(<0.001)	a < b***
	CoT-CL[c]	1.4292	0.49065		a < c**

Table 7
One-way ANOVA of clarity, context accuracy, final answer accuracy, and quality of working across three subjects for GPT-4.

Variable	Subject (n = 120)	Mean	SD	F (p)	Scheffe
Clarity	Chemistry[a]	1.9125	0.26459	8.996	c < b***
	Physics[b]	1.9917	0.09129	(<0.001)	
	Mathematics[c]	1.8417	0.38339		
Context accuracy	Chemistry[a]	1.4667	0.50098	1.809	
	Physics[b]	1.6083	0.79172	(0.165)	
	Mathematics[c]	1.5917	0.56130		
Final answer accuracy	Chemistry[a]	1.7875	0.39248	62.551	c < a***
	Physics[b]	0.7250	0.76655	(<0.001)	
	Mathematics[c]	1.3292	0.94490		b < a***
					b < c***
Quality of working	Chemistry[a]	1.1167	0.31578	110.309	a < c***
	Physics[b]	1.2167	0.77982	(<0.001)	
	Mathematics[c]	1.9792	0.13590		b < c***

correct or incorrect final answer, which was difficult to explain. In a case involving a chemistry answer.

An incorrect significant figure calculation during an intermediate step still resulted in a correct final answer (see Case 4, Chemistry Topic 2, in Table 8). In another case, an incorrect significant figure calculation in an intermediate step led to an incorrect final answer (see Case 3, Chemistry Topic 1, in Table 8). These cases in answer generation were not exceptional but occurred multiple times across three subjects. Similar cases were also found in physics and mathematics. For example, as shown in Case 6, Physics Topic 2 (Table 8), although the multistep answers did not address all the steps correctly, the derivation of a formula was not presented, and GPT-4 with CoT-CL generated a correct final answer. Case 7, Mathematics Topic 1 (Table 8), showed that the quality of working was scored as a 2 and was clear, but the final answer was not correct.

5.2.2. *Domain-specific errors in generated questions and answers*

There were common errors identified across the respective subjects and topics (see Table 9). The most prevalent error across the three subjects was related to significant figure calculations, in particular improper rounding and/or incorrect determination of the number of significant figures. As shown in Table 9, these errors occurred frequently in multistep and final answers. This error pattern was not subject specific but rather calculation type specific.

On the other hand, subject- or topic-specific errors were also observed. In chemistry, a few cases involved redundant information in intermediate steps that were not used in solving the questions (see Case 4, Chemistry Topic 2, in Table 8). In physics, there were multiple errors related to inappropriate notation when dealing with both vertical and horizontal dimensions in a two-dimensional coordinate system. Case 5, Physics Topic 1 (Table 8), provides an example of this. In addition, in physics, formulas generally need to be derived during solving equations and calculation; however, some cases did not include the derivation of formulas (see Case 6, Physics Topic 2, in Table 8). In mathematics, there were many cases involving unrealistic values given the real-life context, such as excessively high or lower prices of items or an unusually large number of objects, which are not typical in everyday life (see Case 8, Mathematics Topic 2, in Table 8).

6. Discussions

This study investigated the quality and accuracy of using LLMs to

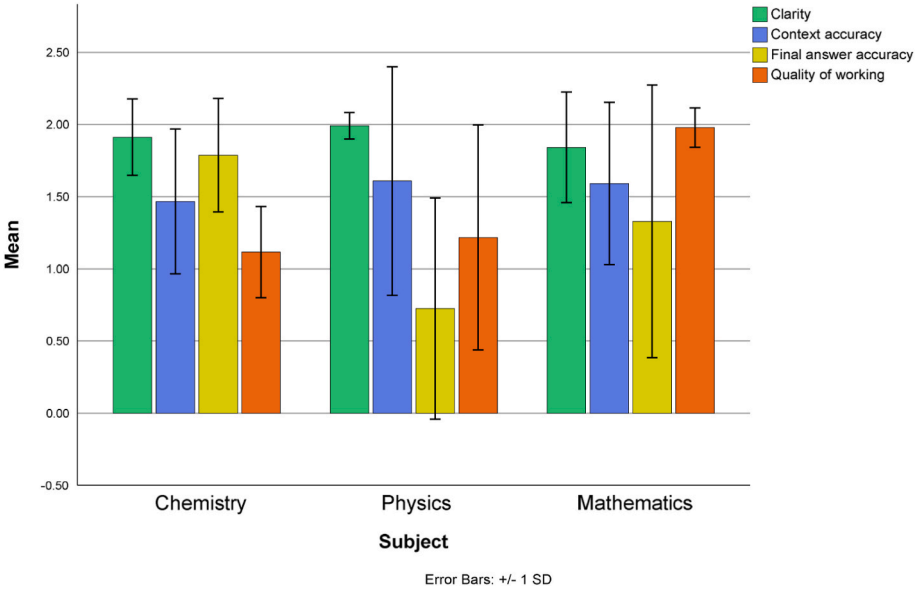


Fig. 3. Comparison across three subjects for GPT-4.

Table 8
Examples having errors in generated answers.

Case	Subject (Topic)	Model (prompting)	Question	Answer	Comments for final answer accuracy (score)	Comments for quality of working (score)
Case 1	Physics (Topic 1)	GPT-4 (CoT)	Q8. A football is kicked with a velocity of 25 m/s at an angle of 45° above the horizontal. It lands in a depression 1.0 m deep. Calculate the velocity of the football just before it lands in the depression.	Calculate the vertical component of the initial velocity (uy): uy = 25 * sin (45°) ≈ 17.68 m/s Using the kinematic equation for vertical motion: vy^2 = uy^2 + 2 * a * sy vy^2 = (17.68)^2 + 2 * (−9.81) * (−1.0) ≈ 329.37 vy ≈ 18.15 m/s Calculate the horizontal component of the initial velocity (vx): vx = 25 * cos(45°) ≈ 17.68 m/s Compute the total velocity just before hitting the bottom: v^2 = vx^2 + vy^2 v^2 = (17.68)^2 + (18.15)^2 ≈ 633.05 v ≈ 25.16 m/s Determine the angle below the horizontal: tan(theta) = vy/vx theta = atan(18.15/17.68) ≈ 45.8° below horizontal	The values of both velocity and angle calculated are wrong, even though workings are correct (0)	Clear, detailed workings (2)
Case 2	Mathematics (Topic 1)	GPT-4 (CoT)	Q1. A store sells three types of lamps: Type A, Type B, and Type C. The total cost of 10 Type A, 15 Type B, and 8 Type C lamps is $1125.40. The total cost of 4 Type A lamps is equal to the total cost of 5 Type B lamps. The total cost of 12 Type C lamps is $38.40 more than the total cost of 8 Type A and 8 Type B lamps. Find the cost of one Type C lamp.	Let the cost (in $) of one of Types A, B, and C lamps be A, B, and C respectively. The equation based on total cost is: 10A + 15B+8C = 1125.40 The equivalence of costs between A and B is: 4A−5B = 0 The comparison of costs involving Type C lamps is: −8A−8B + 12C = 38.40 Solving these equations, we find: A = 33.60 B = 26.88 C = 40.50 Thus, the cost of one Type C lamp is $40.50.	Answer is wrong. Using GC, x = 24.8, y = 27.8, z = 45.0 (0)	Clear (2)
Case 3	Chemistry (Topic 1)	GPT-4 (CoT-CL)	Q20. A 3.25 g sample of a compound used in fireproof coatings, which includes magnesium carbonate, is added to 95 cm^3 of 0.165 mol dm^{-3} aqueous hydrochloric acid. The excess acid is neutralized by 23.00 cm^3 of 0.075 mol dm^{-3} aqueous potassium hydroxide. Calculate the percentage of magnesium carbonate in the fireproof coating.	$MgCO_3 + 2HCl \rightarrow MgCl_2 + H_2O + CO_2$ Amount of initial HCl = 95/1000 x 0.165 = 0.015675 mol Amount of HCl in excess = 23.00/1000 x 0.075 = 0.001725 mol Amount of reacted HCl = 0.015675–0.001725 = 0.01395 mol Amount of $MgCO_3$ = 0.01395/2 = 0.006975 mol Mass of $MgCO_3$ = 0.006975 x (24.3 + 12 + 48) = 0.582 g Percentage of magnesium carbonate = 0.582/3.25 x 100 = 17.9%	Final answer is incorrect because of the second last step. The answer is slightly off. (less than 0.01 difference) (1)	Intermediate answers should be 4–5 significant figures. Final answer should be 3 significant figures (1)
Case 4	Chemistry (Topic 2)	GPT-4 (CoT-CL)	Q1. Using inert electrodes, a current was passed through two beakers containing aqueous silver nitrate and aqueous chromium (III) sulfate respectively and connected in series. After 35 min, 0.083 g of silver was deposited from the first solution. Calculate the mass of chromium deposited from aqueous chromium (III) sulfate.	Mass of Ag deposited = 0.083 g/107.9 g/mol = 0.000769 mol Quantity of charge passed = 0.000769 mol * 96,500 C/mol = 74.16 C Current passed = 74.16 C/(35 * 60 s) = 0.0353 A Moles of Cr deposited = 74.16 C/(3 * 96,500 C/mol) = 0.000256 mol Mass of Cr deposited = 0.000256 mol * 52.0 g/mol = 0.0133 g	Final answer is correct (2)	Intermediate answers should be 4–5 significant figures. There is a rounding off error for Step 2, which should be (0.000769 mol* 96,500 C/mol) = 74.2085 C. Current (A) was not used (redundant information) (1)
Case 5	Physics (Topic 1)	GPT-4 (CoT-CL)	Q3. A ball was kicked with a velocity of 22 m/s at an angle of 41° above the horizontal. It landed into a depression 2.9 m deep.	vx = 22 * cos(41) = 16.60 m/s vy = 22 * sin(41) = 14.43 m/s vy^2 = (14.43)^2 + 2 * −9.81 * −2.9 = 265.73 vy = sqrt(265.73) = 16.30 m/s	It is unclear what Python uses to calculate the intermediate value of vy^2, which is slightly off. This causes the final answers to	The same notation is used to denote both initial vertical and final vertical velocity (1)

(*continued on next page*)

Table 8 (*continued*)

Case	Subject (Topic)	Model (prompting)	Question	Answer	Comments for final answer accuracy (score)	Comments for quality of working (score)
				v = sqrt(16.60^2 + 16.30^2) = 23.27 m/s theta = atan(16.30/16.60) = 44.47° below horizontal	be slightly off when expressed in 4 s.f. (1)	
Case 6	Physics (Topic 2)	GPT-4 (CoT-CL)	Q1. A conical pendulum consists of a bob with a mass of 1.63 kg tied to a string that is 1.59 m long. It swings in a circular motion at a constant speed, making an 18.18° angle with the vertical. Calculate the speed of the bob.	Calculate the radius: r = 1.59 * sin (18.18°). Use the formula: v = sqrt(tan (18.18°) *r * 9.81). The speed is approximately 1.26 m/s.	Final velocity is correct (2)	Quote formula to find velocity without deriving (1)
Case 7	Mathematics (Topic 1)	GPT-4 (CoT-CL)	Q8. An electronics store stocks three types of gadgets: Type P, Type Q, and Type R. The total cost of 12 Type P, 17 Type Q, and 7 Type R gadgets is $1235.93. The total cost of 2 Type P gadgets is equal to the total cost of 3 Type Q gadgets. The total cost of 13 Type R gadgets is $43.18 more than the total cost of 9 Type P and 12 Type Q gadgets. Find the cost of one Type R gadget.	Let P, Q, R represent the costs of Types P, Q, and R gadgets, respectively. We have the equations: 12P + 17Q + 7R = 1235.93 2P − 3Q = 0 −9P − 12Q + 13R = 43.18 Solving these gives: P = 37.33, Q = 24.89, R = 52.14 Answer: The cost of one Type R gadget is $52.14.	Answers are correct, but R should round off to 52.13 instead (1)	Clear (2)
Case 8	Mathematics (Topic 2)	GPT-4 (CoT-CL)	Q16. A bike shop sells three types of items: Type Mountain bikes, Type Road bikes, and Type Electric bikes. The total cost of 15 Type Mountain bikes, 18 Type Road bikes, and 11 Type Electric bikes is $1124.85. The total cost of 4 Type Mountain bikes is equal to the total cost of 3 Type Road bikes. The total cost of 16 Type Electric bikes is $42.36 more than the total cost of 15 Type Mountain bikes and 18 Type Road bikes. Find the cost of one Type Electric bikes.	Let the cost of one Type Mountain bikes, Road bikes, and Electric bikes be X, Y, and Z respectively. 15X + 18Y + 11Z = 1124.85, 4X - 3Y = 0, −15X - 18Y + 16Z = 42.36. Solving, X = 16.65, Y = 22.20, and Z = 43.23. The cost of one Type Electric bikes is $43.23.	Accurate (2)	Clear (2)

Table 9
Error patterns in questions and answers.

Subject	Error pattern
Chemistry	● Errors in significant figure calculations ● Redundant information in intermediate steps
Physics	● Errors in significant figure calculations ● Errors in notations for vertical axis ● Missing a formula deriving
Mathematics	● Errors in significant figure calculations ● Unrealistic values (e.g., prices of items) of real-life examples

automatically generate questions and answers. Based on the quantitative and qualitative findings of this study, we discuss (1) areas of agreement with previous studies, (2) areas not fully aligned with previous studies, and (3) extensions as contributions to AIG research through this study.

6.1. Agreement 1: GPT-4 outperformed GPT-3.5 for AIG in general

One of the common findings of recent studies in the area of AIG using LLMs is that GPT-4, a newer LLM, outperformed GPT-3.5, the previous model (Indran et al., 2024). This study also supports this trend in general, as we compared GPT-3.5 and GPT-4 for AIG across three different prompting approaches (see Table 1 and Fig. 1). In particular, the finding highlights that the quality of items generated by GPT-4 using SP was considerably higher than that of GPT-3.5 in three out of four aspects: clarity of questions, context accuracy of questions, final answer accuracy, and quality of working for answers (see Fig. 1). This result underscores the necessity of using the most recent LLMs for AIG to reliably generate more quality items. Considering that recent studies using GPT-3.5 have reported that the quality of automatically generated items was satisfactory (e.g., Cheung et al., 2023; Sayin & Gierl, 2024), simple items can be generated with sufficient quality using CoT prompting. The findings revealed that the average scores for clarity of questions and context accuracy of questions were comparable between GPT-3.5 and GPT-4. This result might be attributed to the fact that automatically generated questions in previous studies involved only several sentences, making the task relatively straightforward.

However, generating items that entail a more complex process, such as generating multiple steps of answers in our study, there was still a challenge with GPT-3.5. GPT-3.5 is freely available to the public, and despite its ability to generate items of decent quality in particular areas, the quality is not assured without thorough validation and fine tuning. Thus, it is advisable to use the most recent LLMs, which have more parameters, to secure the quality of AIG in general. Recently, newer models, such as GPT-4o and OpenAI o1-preview, have been released and have been widely used. Although the benchmarks indicate that these

newer models outperform GPT-3.5 in general (e.g., OpenAI, 2024a), more investigations in the area of AIG in STEM subjects should be conducted.

6.2. Agreement 2: The use of CoT prompting in generating quality questions

Given the well-recognized effectiveness of the CoT prompting approach (Cohn et al., 2024), there has been an increasing utilization of CoT for AIG (Xie et al., 2024). The findings of this study are aligned with recent studies' findings in general. As shown in Table 6, the average scores generated by SP, which involved contextual information for each subject, were considerably lower than those of CoT across GPT-3.5 and GPT-4. In particular, the score gap in Context accuracy between SP and CoT was the largest. Context accuracy pertains to the realistic and factual aspects of generated questions, such as whether the generated values, such as mass, are representative of everyday life examples. It means that CoT is a useful and powerful tool to generate reasonable contexts of questions.

When employing CoT, both GPT-3.5 and GPT-4 generated reasonable values that aligned with common usages, while SP produced unusual numbers (see Table 6). This finding suggests that providing structured input (Circi et al., 2023) with contextual information, such as subjects and levels of teaching, and following a specific input scheme is critically important for generating better quality items. Interestingly, when generating questions that followed sample questions with CoT using LLMs, there was no significant difference between GPT-3.5 and GPT-4. This finding may imply that the proper use of the CoT approach is more crucial than the selection of LLMs in cases with relatively simpler structures that are relatively straightforward.

However, this was not always the case for more complex tasks. One of the findings of this study also showed that generating answers involving multiple steps resulted in lower item quality with the CoT approach. This may raise a challenge for the CoT approach. This is discussed in the following section.

6.3. Disagreement 1: limitation of the CoT prompting in generating multistep answers

Many recent studies have reported the significant effectiveness of the CoT approach in prompting (e.g., Lee et al., 2024; Wei et al., 2022). However, the findings of this study may reveal a limitation of the CoT approach in universally generating quality items, as shown in the penultimate row in Tables 6 and in Fig. 1. One of the experiments of this study involved generating answers to questions that required multistep calculations, which might be more complex than generating the questions themselves. The result showed lower average scores: 1.0000 out of 2.0 for GPT-4 with the CoT approach, which was not significantly different from the score 0.9958 of SP (see Table 6). The average scores of this measure were not satisfactory: This was also the exception among the four measures used to evaluate the quality and accuracy of the generated questions and answers. It was interesting to note that the CoT approach performed well, scoring 1.6667 out of 2.0 in terms of quality of working when generating multiple steps. This indicates that GPT-4 was capable of generating logically sound multistep answers using the CoT approach, which is its most compelling aspect. However, there may be limitations to the CoT prompting approach in ensuring correct calculations within multistep answers (see Table 8).

One very recent study also reported that LLMs face challenges in generating answers to complex mathematical questions at an undergraduate level (Satpute et al., 2024). Satpute et al. suspected that generating mathematical expressions may involve unwritten rules and assumptions with precise knowledge, which is not similar to other natural language tasks. Likewise, the investigations conducted in this study, which involved more complex and multiple steps of calculations in physics, chemistry, and mathematics, may face similar challenges for LLMs in AIG, partially due to the probabilistic reasoning nature of LLMs (Nafar et al., 2024). Although there could be better prompting with the CoT approach, it is clear that this type of task demands more fine tuning to achieve an adequate quality of items.

6.4. Disagreement 2: Identification of a dual processing at intermediate steps of AIG

Many studies have generally stated that LLMs reason or explain things based on given prompts (e.g., Balhorn et al., 2024). They may have assumed that the answers generated in LLMs using a prompt are logically interconnected. However, we found that dual processing might be involved in generating multistep answers. One process may involve arriving at the final answer using the information provided in the question, while the other involves displaying multiple steps externally, guided by prompts. These two separate processes do not always seem to align. We observed cases in physics and chemistry answers where the final answer was correct, but the intermediate steps were inaccurate (see Table 8). These instances mean that multistep answers generated by LLMs do not necessarily reflect the internal processing used to solve the question.

In particular, the observed cases mostly involved the calculation of significant figures. Significant figures are a basic element in science, taught in science subjects due to measurement uncertainty (Wan, 2023). The answer is usually incorrect if the process does not follow the rules for significant figures. Therefore, students are taught to solve science questions with the notion of significant figures, while some mathematical questions involve proper rounding. However, some cases in this study produced accurate final answers despite incorrect intermediate steps represented in terms of the number of significant figures. Another example from physics also demonstrates this dual processing (see Case 5, Physics Topic 1, in Table 8). Although there was a missing step in linking centripetal force with gravitational components and the use of $\tan(\theta)$, the formula was still used to calculate the final velocity. GPT-4 may have followed the logic of sample multistep answers to calculate the final answer successfully, even when there was a missing connection in the multistep answer. These cases may suggest dual processing at intermediate steps in LLMs.

Since the mechanism of AIG using LLMs is not sufficiently transparent or explainable (Chang et al., 2024), scrutinizing the intermediate steps is challenging. With this in mind, this issue raises two approaches to avoiding problems possibly caused by dual processing: (1) creating two different prompts, one to represent multistep answers and one to solve questions, and (2) improving the quality of prompts to handle significant figures in calculations. For the first point, it would be possible to develop optimal prompting. Regarding the second point, we suggest incorporating coding language into the CoT approach, discussed in the following section.

6.5. Extension 1: Incorporating coding language into CoT prompting

The findings of this study revealed that CoT-CL outperformed SP and was comparable with CoT. Intriguingly, the average scores of final answer accuracy using CoT-CL were significantly higher than with SP and CoT (see Table 6). This generation required a multistep process involving the calculation of significant figures, which was more complex, to obtain a final answer. We found cases where the final answers were incorrect due to incorrect significant figure digits in the intermediate steps when using the SP and CoT approaches. However, CoT-CL mitigated this intermediate issue through an additional verification process involving the Code Interpreter function embedded in GPT-4. We hypothesize that this improvement was possible as the process was logically sequenced and explicitly verified step by step. This suggests that GPT-4 may have limitations in solving STEM subject questions that involve a multistep process of calculating equations with multiple variables without further explicit logical prompts. For example, as shown in

Cases 1 and 2 (Table 8), the composition of equations and their sequential process were clear, but the calculated values were incorrect. These issues were significantly mitigated when using the CoT-CL approach, which achieved a score (1.8458) significantly higher than that of the CoT approach (1.0000; see Table 6). The reason the CoT approach improves the quality of complex reasoning significantly is that it gives instructions to LLMs to follow a particular logical sequence (Wei et al., 2022). Likewise, the CoT-CL approach may further enhance this process, particularly in solving multiple equations with many variables, including the calculation of significant figures, making it more logical and resulting in more promising and applicable utilization for AIG in STEM subjects.

Similar to the usability of the CoT approach, the application of the CoT-CL approach is straightforward. The prompt 'Please write Python code to emulate the mark scheme, and run the code to ensure that you get the same answer. Otherwise, rewrite and rerun the code' was used for this study. No expert knowledge for coding language is required to employ this approach; it simply involves prompting to run the code to ensure the steps are correct, which can be easily done by STEM subject educators.

6.6. Extension 2: Conceptual structure and domain-specific prompting rules

It is clear that particular prompts are essential to generate questions and answers that are more accurate and realistic. Through developing these prompts and analyzing the results, we were able to establish STEM disciplinary prompting rules for AIG. We found that every question had (1) key concepts as patterns structuring the question and (2) unwritten rules in STEM subjects. Thus, to generate more accurate and high-quality items using LLMs, it was necessary to prompt according to these patterns and state the rules specific to STEM subjects. When we applied these patterns in generating similar questions, the quality of the questions became consistent across the three subjects studied.

The key conceptual patterns in the questions were: chemical reactions for chemistry, similar motion scenarios for physics, and systems of linear equations or permutation and combination problems for mathematics. These patterns may be deeply related to the logical structure of key concepts in each subject, particularly in calculative questions. Hence, it would be imperative to clearly state the concepts being used for generating questions. In addition, it was necessary to prompt domain-specific rules for each subject and topic (see Table 10). For chemistry, including prompts about ratios and percentages was critical for generating realistic questions. In physics, specifying the plane of motion (e.g., horizontal circular motion) was imperative; otherwise, the result was unrealistic or redundant values in the question such that there was either no solution or contradicting multiple solutions. For mathematics, prompting relevant real-life examples involving quantities, prices, and values was required; otherwise, unrealistic numbers and prices of items were generated.

The identification of key concepts in questions and subject-specific prompting rules suggests a useful approach to developing prompts for AIG using LLMs. Given that LLMs facilitate human-like conversations (Kasneci et al., 2023), there will be considerable attempts to generate assessment items automatically. Considering conceptual structure and domain-specific rules to develop prompts may be helpful to reducing the effort and time needed for AIG in STEM subjects.

Table 10
Subjects' and topics' key conceptual patterns in questions and prompting rules.

	Conceptual structure in questions	Subject/topic prompting rules
Chemistry	Same chemicals for Topics 1 and 2	Reasonable ratio (e.g., percentage)
Physics	Similar motion for Topics 1 and 2	Coordination of the motions (e.g., with the vertical axis, having a depression)
Mathematics	System of linear equations for Topic 1 Permutations and combinations for Topic 2	Real-life relevant context

7. Limitations of this study

This study systematically investigated the accuracy and quality of questions and their multistep answers generated by GPT-3.5 and GPT-4 using three different prompting approaches, focusing on three STEM subjects at the pre-university level. However, this investigation had several limitations that should be carefully considered when interpreting and generalizing the findings. First, the findings of this study would be applicable only to the three subjects—chemistry, physics, and mathematics—at the pre-university level and may not extend to other subjects or university-level questions. As shown in the findings of this study, GPT-4's capabilities in generating accurate multistep answers varied across the three subjects, possibly due to their differing conceptual structures, which suggests that its accuracy may also vary in other subjects. In addition, as reported by Balhorn et al. (2024), questions requiring more complex reasoning tend to result in lower accuracy for answers generated by LLMs. Likewise, generating less-structured questions that involve complex reasoning may pose challenges for LLMs in producing high-quality questions and answers. Second, the accuracy and quality of generated questions and answers may vary when using newer models, such as GPT-4o, OpenAI o1, Claude 3.5, and Llama 3.1, or when employing fine-tuning methods. For example, OpenAI o1 has demonstrated superior performance in mathematical tasks and complex reasoning compared to other LLMs including older versions (OpenAI, 2024b). Newer models are expected to perform better in generating higher-quality questions. While newer LLMs and fine-tuning approaches will continue to evolve, we believe the findings of this study—such as the benefits of incorporating coding language into CoT and identifying questions' conceptual structures and domain-specific prompting rules in STEM subjects—still provide valuable contributions to this area of research.

8. Conclusion and implications

Along with the advancement of LLMs, AIG research has been increasingly gaining attention due to the potential use of these technologies (Circi et al., 2023). This trend suggests a need for further investigation into how to enhance the quality and reliability of automatically generated items. This need is imperative, given the development of prompting techniques and the current capabilities of LLMs in specific contexts in different domains, of different question types and mastery levels. Although there have been many studies conducted to investigate capabilities of LLMs (e.g., Cheung et al., 2024), most studies have focused on MCQs. With this in mind, this study systematically examined how well two LLMs, GPT-3.5 and GPT-4, generated questions and multistep answers for Year 12 students aged 17–18 in the STEM subjects of chemistry, physics, and mathematics in terms of conceptual accuracy and quality of working. To investigate the accuracy and quality of the generated items, we compared SP, CoT, and CoT-CL across the three different STEM subjects. Overall, AIG in STEM subjects showed substantial quality in both questions and answers when employing the CoT approach for both GPT-3.5 and GPT-4 models with structured input. In particular, the use of GPT-4 with the integration of the CoT-CL approach showed very high quality for generating questions and substantial accuracy and quality for generating multistep answers across the three different subjects. On the other hand, this study also revealed domain-specific prompting rules for AIG, which is one of the essential aspects of prompt engineering.

The findings of this study have significant implications for AIG in STEM subjects. First, this study reinforces the direction of AIG research applying LLMs. Given the promising findings, teachers can be

encouraged to draft questions and multistep answers by prompting LLMs. Problem-solving activities have been one of the key components of many science and mathematics classes, as students are able to integrate their knowledge by engaging in more active cognitive processes during these activities (Mayer, 2002; McDermott, 2001). In particular, since types of questions requiring multistep answers with calculations are commonly incorporated into STEM classes to enhance students' conceptual understanding and ability to apply their knowledge, teachers drafting questions supported by AIG can be beneficial in that it can enable them to generate questions and answers more efficiently, thereby securing more time to support students' learning (Circi et al., 2023). Given that LLMs have demonstrated varying capabilities in generating content depending on different domain-specific knowledge (Balhorn et al., 2024; OpenAI, 2023), and that each subject may have a different typical conceptual structure for questions, as found in this study, it will be necessary to investigate approaches, including prompting, that can broaden the scope to other STEM subjects, such as biology and computer science. To this end, further investigation could begin with the identification of typical conceptual structures and specific prompting rules (see Table 10).

Second, this study suggests practical implications to generating quality items in STEM subjects, promoting more active utilization of LLMs for AIG. In particular, this study provided practical cases, including prompting techniques such as providing contextual information, few-shot prompting, CoT, and the integration of coding language into CoT, which will be useful empirical approaches for future investigations. As discussed in the literature review, the accessibility and adaptability of LLMs have been significantly improved for the general public, including those who are not computer language experts. This suggests that many educators interested in AIG may find it relatively easy to learn how to generate questions and answers using LLMs. Based on the findings of this study, derived from both quantitative and qualitative analyses, we recommend the following points for AIG using LLMs in STEM subjects.

(a) Provide sufficient contextual information, such as subject, topic, level of teaching, and the context of the questions, to generate relevant questions.
(b) Include sample questions and answers.
(c) Prompt the model to follow the reasoning of the sample questions and answers step by step.
(d) Prompt the model to write code and verify the process.
(e) Verify the generated questions and answers with subject-matter experts, focusing on numerical calculations and the use of realistic values.
(f) Fine-tune the prompts using validation from subject-matter experts, incorporating domain-specific prompt rules where identifiable.

These recommendations outlined in points (a) to (f), are illustrated in Table 3 with specific examples. Points (e) and (f) were specifically drawn from the findings of this study. While LLMs can generate questions and answers of comparable quality, it remains necessary for subject-matter experts to validate AI-generated content. During this validation process, educators may identify domain-specific prompt rules to tailor the generated items, streamlining future item generation processes. Recently, AIG using LLMs has become even more convenient, as individuals can now develop their own customized generative AI chatbots and share them with others. For example, Kıyak and Kononowicz (2024) developed a sharable case-based MCQ generator using ChatGPT that incorporates embedded prompts to help other educators easily generate more targeted questions focused on particular cases. Similarly, educators can begin developing customized generative AI chatbots for AIG and share them with educators, potentially increasing scalability while adhering to the recommended points outlined above. Through the development of AIG using LLM prompting, we believe that educators will be able to use their time and effort more effectively for their teaching.

Lastly, educators should carefully consider ethical issues related to using AI-generated content for assessments, in particular in high-stakes exams such as summative assessments. Most importantly, educators should take responsibility for generating and reviewing questions and answers used in assessments. While using LLMs can produce questions of comparable accuracy and quality with proper prompting techniques, as shown by this study, and can significantly reduce time and effort (Cheung et al., 2024), AI-generated answers are not always correct. This is because LLMs may generate hallucinations (Mishra et al., 2023) and may struggle with logical and mathematical tasks (Satpute et al., 2024) due to the probabilistic nature of content generation (Nafar et al., 2024; Xu, Jain, & Kankanhalli, 2024). In addition, LLMs may involve biases related to gender, culture, and other factors due to the biases present in their pre-training data (Zhang & Tur, 2024), which may be more prominent in non-STEM subjects. For example, in language education, there could be potential validity issues related to establishing fair constructs for assessment (Lin & Chen, 2024), as language testing typically involves socio-cognitive frameworks (Weir, 2005). This suggests that AIG using LLMs, given their nature, may require different ethical considerations depending on the subject area. Along with this, it is critically important for teachers to use AI-generated content for assessment responsibly. Professional development programs could support teachers not only by helping them understand prompting techniques for AI-generated questions and answers, but also by discussing ethical guidelines tailored to using AI in assessment contexts. By addressing potential ethical issues and taking steps to mitigate them, the use of LLMs in generating assessment items could become a more promising tool for supporting teachers with evolving prompting approaches.

CRediT authorship contribution statement

Kuang Wen Chan: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Farhan Ali:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Joonhyeong Park:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kah Shen Brandon Sham:** Writing – review & editing, Methodology, Formal analysis. **Erdalyn Yeh Thong Tan:** Writing – review & editing, Methodology, Formal analysis. **Francis Woon Chien Chong:** Writing – review & editing, Methodology, Formal analysis. **Kun Qian:** Writing – review & editing, Formal analysis. **Guan Kheng Sze:** Writing – review & editing, Formal analysis.

Data availability statement

The prompts used in this study are available in the supplementary document.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2024.100344.

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